AZAZ UR REHMAN

AI Engineer

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience in building AI-powered systems and applications. Recently completed a role as Associate AI Engineer at Cware Technologies, where I worked on LLM integrations, RAG pipelines, NLP solutions, and backend API development using Python and FastAPI. Graduated from NUTECH with a degree in Computer Science. Have also worked on Machine Learning, Deep Learning, and Computer Vision projects across multiple internships.

EDUCATION

Bachelor of Science in Computer Science
National University of Technology (NUTECH) | Nov 2021 – Jun 2025 | Islamabad | GPA: 3.05

PROFESSIONAL EXPERIENCE

Associate Software Developer | *NUTECH Innovation Center (NIC)* | *Mar 2026 – Sep 2026* | *Islamabad*

- Developed and enhanced mobile applications using modern Android development technologies (Kotlin, XML), including offline map integration for plotting coordinates (Eastings and Northings).
- Implemented and integrated digital and vector maps, including map rendering, map layers, and tile-based mapping solutions.
- Worked with Heltec modules for offline data communication between host and station devices, building the companion Android app for this communication.
- Contributed to the development of an AI-based weather prediction system, covering data preparation, model development, training, and evaluation using historical weather data.

Associate AI Engineer | *Cware Technologies* | *Nov 2025 – May 2026* | *Islamabad (Remote)*

- Designed and developed AI-powered backend systems using Python, FastAPI, and PostgreSQL, enabling intelligent data processing and automation.
- Built and integrated LLM-based workflows using OpenAI API and LangChain, including RAG pipelines with FAISS vector databases for semantic search and retrieval.
- Implemented NLP techniques for information extraction, text classification, and document understanding within production-grade APIs.
- Built the SIMAPP multi-industry management platform backend with AI-enhanced features for Telecommunication, Event Management, and Logistics.
- Delivered the CardTopup and GitHub Automation projects, incorporating intelligent automation and API integration.

Associate Deep Learning Engineer | *IT Solera* | *Jun 2025 – Aug 2025* | *Islamabad (Remote)*

- Built deep learning models using TensorFlow and PyTorch for computer vision applications.
- Performed image preprocessing, data augmentation, model training, and performance evaluation.
- Worked with advanced CNN architectures including ResNet and VGG-17/19 for image classification tasks.

AI & ML Intern | *National Disaster Management Authority (NDMA)* | *Jul 2024 – Aug 2024* | *Islamabad*

- Assisted in AI/ML research, data analysis, and technical reporting for disaster management scenarios.
- Developed analytical reports using real-world disaster data to support risk assessment decisions.
- Explored and applied machine learning techniques for predictive decision-making and risk analysis.

TECHNICAL SKILLS

AI & LLM: OpenAI API, LangChain, RAG Pipelines, FAISS, Prompt Engineering, Whisper

Machine Learning: Scikit-learn, XGBoost, Random Forest, SVM, Decision Trees, Feature Engineering

Deep Learning: TensorFlow, PyTorch, CNN, ResNet, DenseNet, EfficientNet, AlexNet, LeNet, FaceNet

Computer Vision: YOLO, OpenCV, Image Preprocessing, Data Augmentation, Object Detection

NLP: Text Classification, Information Extraction, Sentiment Analysis, Speech-to-Text (Whisper)

Backend Development: Python, FastAPI, Flask, REST API Development, JWT Authentication

Databases: PostgreSQL, MySQL, SQLite, Alembic, JSONB, FAISS, Prisma

Programming Languages: Python, C++, Java, C#

Version Control: Git, GitHub

PROJECTS

AI-Based Weather Prediction System | *Completed*
Built a weather prediction system: downloaded and processed ERA5 reanalysis weather data from an online source using Python, and trained a SimVP (Simple Video Prediction) model from scratch on the dataset to forecast short-term weather conditions.
Technologies: Python, ERA5 Dataset, SimVP, Deep Learning

WordPress Product Image Automation Tool
Built a local Windows tool that automates WooCommerce product image listing: it collects candidate images with a headless browser, uses Claude (Anthropic API) to pick the best 4–5, processes them, and uploads them as the main image and gallery, with human approval before publishing.
Technologies: Python, FastAPI, SQLAlchemy, SQLite, Playwright, Pillow, Anthropic API, React, Vite, TypeScript, Tailwind CSS

AI Resume Analyzer | *Jun 2026*
Built an AI-powered Resume Analyzer using LangChain and LLMs to evaluate resumes against job descriptions. Implemented ATS scoring, skill gap analysis, interview question generation, and personalized improvement suggestions. Enabled users to upload resumes, receive detailed feedback, and download enhanced versions.
Technologies: Python, LangChain, OpenAI API, FastAPI, SQLite, FAISS

MedScribe AI – Voice Medical Assistant | *Jul 2025*
Developed an AI-powered medical voice assistant that allows patients to describe their symptoms by voice. Used OpenAI Whisper to convert speech into text, applied NLP to extract medical information, and generated structured medical notes through FastAPI-based backend services.
Technologies: Python, Whisper, OpenAI API, FastAPI, NLP

Vehicle Detection & Counting System | *Mar 2026*
Developed a real-time vehicle detection and counting system using YOLO and OpenCV for traffic monitoring. Implemented object tracking to count vehicles from live video streams and generate traffic statistics.
Technologies: Python, YOLO, OpenCV

Cheating Detection System (Deep Learning + CatBoost) | *Jul 2025*
Built a cheating detection system that extracts image features using EfficientNetB0/B3 and classifies them with a CatBoost classifier, achieving ~90% testing accuracy. Applied strong data augmentation for better generalization and evaluated performance using classification reports, confusion matrices, and ROC-AUC. Designed with future potential for real-time webcam-based detection.
Technologies: Python, TensorFlow/Keras, EfficientNetB0/B3, CatBoost, OpenCV

Violence Detection System (Transfer Learning – ResNet50) | *Jun 2025*
Developed a violence detection system by merging multiple datasets (Real Life Violence Dataset and Surveillance Camera Fight Dataset) into a single comprehensive dataset, extracting frames, and applying strong data augmentation. Trained a ResNet50 transfer-learning model achieving ~90% accuracy, and validated its generalization by testing on completely unseen video data outside the training and validation sets.
Technologies: Python, TensorFlow/Keras, ResNet50, OpenCV, Transfer Learning

SIMAPP – Multi-Industry Management Platform | *Nov 2025 – May 2026*
Designed and developed the complete AI-enhanced backend for a multi-industry platform supporting Telecommunication, Event Management, and Logistics, including secure authentication, QR/Barcode generation, and intelligent workflows.
Technologies: Python, FastAPI, PostgreSQL, JWT, Alembic, JSONB, REST APIs

Diabetic Retinopathy Detection (Final Year Project) | *May 2025*
Developed a deep learning system to detect diabetic retinopathy from retinal images. Evaluated CNN architectures (DenseNet, ResNet, EfficientNet, AlexNet, LeNet) and proposed a hybrid DenseNet + XGBoost classifier to improve prediction accuracy.
Technologies: Python, TensorFlow, DenseNet, XGBoost, OpenCV

CERTIFICATIONS

AI and Data Science Certification – NAVTTC, Pakistan